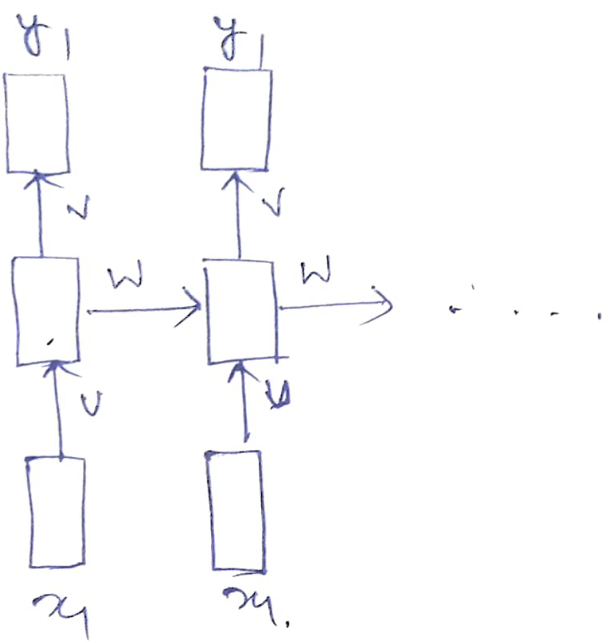
RNN – Backpropagation through Time. (1)

Parameters:



$x_i \in R^n$ (n-dimensional Input)

$s_i \in R^d$ (d-dimensional state)

$y_i \in R^k$ (k-classes)

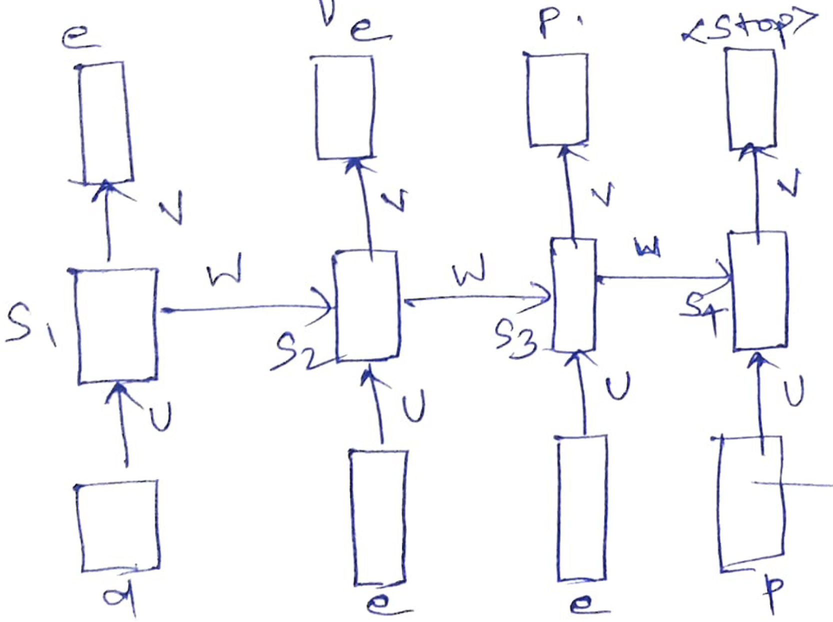
$U \in R^{n \times d}$

$V \in R^{d \times k}$

$W \in R^{d \times d}$.

Train the Network: → Task of Auto prediction (or) Completion

* For simplicity assume, there are only 4 characters in our vocabulary (d, e, p, <stop>)
* At each time step, to predict one of these 4 characters.
* Output function – Softmax.
  Loss function – Cross entropy.



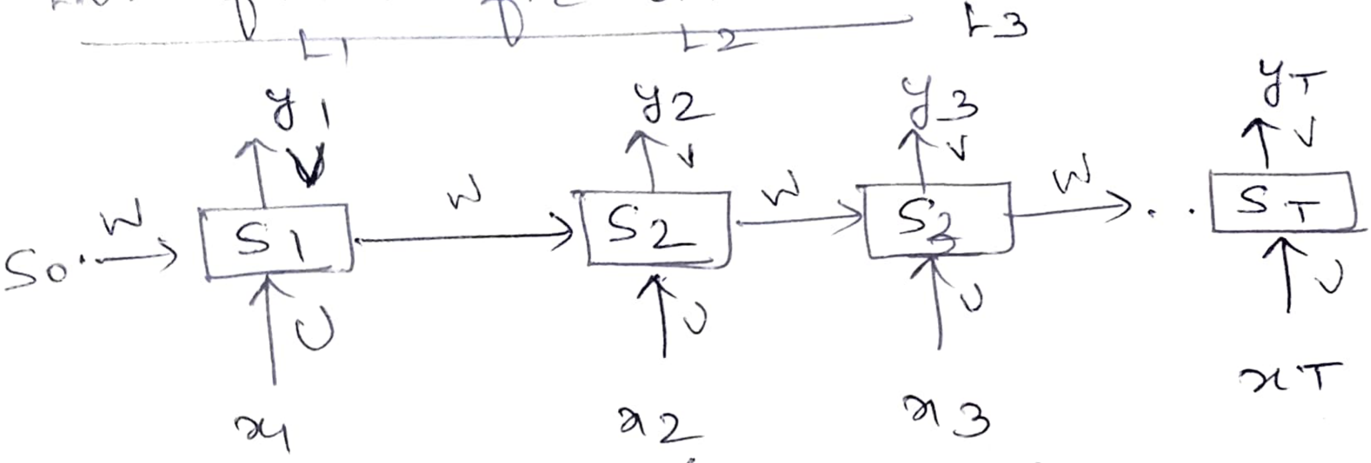
Initialize U, V, W randomly and the N/W predict the probabilities.

Predicted		Predicted $\hat{y}$	True
time step 1:	d	0.2	0
	e	0.7 ✓	1
	p	0.1	0
	Stop	0.1	0
t=2	d	0.2	0
	e	0.7 ✓	1
	p	0.1	0
	Stop	0.1	0
t=3	d	0.2	0
	e	0.1	0
	p	0.7 ✓	1
	Stop	0.1	0
t=4	d	0.2	0
	e	0.1	0
	p	0.1	0
	Stop	0.7	1

(1) Total loss made by the model?

(2) How do backpropagate the loss and update the parameters ($\theta = \{U, V, W, b, c\}$) of the network?

Loss function for a RNN.



$$L = \sum_{i=1}^{T} L_T$$ Sum of Intermediate losses.

↳ Cross entropy – classification

⇒ $-\log(y_{tc})$

↳ Least squares – Regression

Back propagation through time (BPTT)

$$S_t = \phi(W_{xs} x_t + W_{ss} S_{t-1}) = \phi(U x_t + W S_{t-1})$$

$$y_t = O(W S_t) = O(V S_t)$$

W, U, V donot change over time

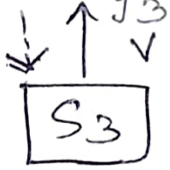
Need to find Gradients:

$$\frac{\partial L}{\partial V}, \frac{\partial L}{\partial W}, \frac{\partial L}{U} \qquad L = \sum_{i=1}^{T} L_t$$

$$\frac{\partial L_3}{\partial V} = \frac{\partial L_3}{\partial y_3} \frac{\partial y_3}{\partial V}$$ [chain rule]

$$= \underset{\text{loss function}}{\downarrow} \cdot S_3 = \dots$$

$$\frac{\partial L_3}{\partial w} = \frac{\partial L_3}{\partial y_3}\,\frac{\partial y_3}{\partial S_3}\,\frac{\partial S_3}{\partial w}.$$



$$= \text{loss function} \cdot V \cdot \frac{\partial S_3}{\partial w}$$

$$S_3 = \phi(\underbrace{U x_3 + W S_2}_{z_3})$$

$$\frac{\partial S_3}{\partial v} = ?$$

$$\frac{\partial S_3}{\partial w} = \frac{\partial \phi}{\partial z_3}\,\frac{\partial z_3}{\partial w}$$

$$= \frac{\partial \phi}{\partial z_3}\left[U\frac{\partial x_3}{\partial w} + S_2(1) + W\frac{\partial S_2}{\partial w}\right]$$

$$= \frac{\partial \phi}{\partial z_3}\left[U\overset{\text{zero}}{\cancel{\frac{\partial x_3}{\partial w}}} + S_2 + W\frac{\partial S_2}{\partial w}\right]$$

$$\frac{\partial S_2}{\partial w} = ? \qquad S_2 = \phi(U x_2 + W S_1)$$

$$\therefore \frac{\partial S_2}{\partial w} = \frac{\partial \phi}{\partial z_2}\left[S_1 + W\frac{\partial S_1}{\partial w}\right]$$

$$\frac{\partial S_1}{\partial w} = \frac{\partial \phi}{\partial z_1}\left[S_0 + W\frac{\partial S_0}{\partial w}\right]$$

$$\therefore \frac{\partial L_3}{\partial w} = \frac{\partial L_3}{\partial y_3}\cdot\frac{\partial y_3}{\partial S_3}\left[\frac{\partial \phi}{\partial z_3}\left(S_2 + W\frac{\partial S_2}{\partial w}\right)\right]$$

$$\frac{\partial L_3}{\partial U} = \frac{\partial L_3}{\partial y_3}\,\frac{\partial y_3}{\partial S_3}\,\frac{\partial S_3}{\partial U}.$$

$\frac{\partial L_3}{\partial y_3} \downarrow$ loss function

$\frac{\partial y_3}{\partial S_3} \downarrow$ V

$\frac{\partial S_3}{\partial U}$ $\quad S_3 = \phi(\underbrace{W S_2 + U x_3}_{Z_3})$

$$\frac{\partial S_3}{\partial U} = \frac{\partial \phi}{\partial Z_3}\,\frac{\partial Z_3}{\partial U}$$

$$= \frac{\partial \phi}{\partial Z_3}\left[U(1) + \frac{\partial W S_2}{\partial U_3}\right]$$

$$= \frac{\partial \phi}{\partial Z_3}\left[U + \frac{\partial W S_2}{\partial U}\right]$$

$$\therefore \frac{\partial W S_2}{\partial U} = W\frac{\partial S_2}{\partial U} + S_2\cancel{\frac{\partial W}{\partial U}}$$

zero.

$$= W\left[\frac{\partial \phi}{\partial Z_2}\cdot\frac{\partial Z_2}{\partial U}\right]$$

Not zero.

$$\frac{\partial L_3}{\partial U} = \frac{\partial L_3}{\partial y_3}\,\frac{\partial y_3}{\partial S_3}\left[\frac{\partial \phi}{\partial Z_3}\left[U + W\frac{\partial S_2}{\partial U}\right]\right]$$